
Social Networks and their Analysis

HOMOPHILY, SOCIAL INFLUENCE, AND AFFILIATION

BY: IVETA MRAZOVÁ



Social Networks and their Analysis

Influence Modeling

- *Linear Threshold Model*
- *Independent Cascade Model*
- *Influence vs. Correlation*

Homophily, Social Influence, and Affiliation

- Homophily Test
- Selection and Social Influence
- Affiliation Networks
- Schelling Model of Segregation
- Positive and Negative Relationships

Homophily, Social Influence, and Affiliation

Motivation:

- “Birds of a feather flock together.”
- Your friends are more similar to you in age, race, interests, opinions, etc., than a random collection of individuals
- **Homophily**: the principle that we tend to be similar to our friends

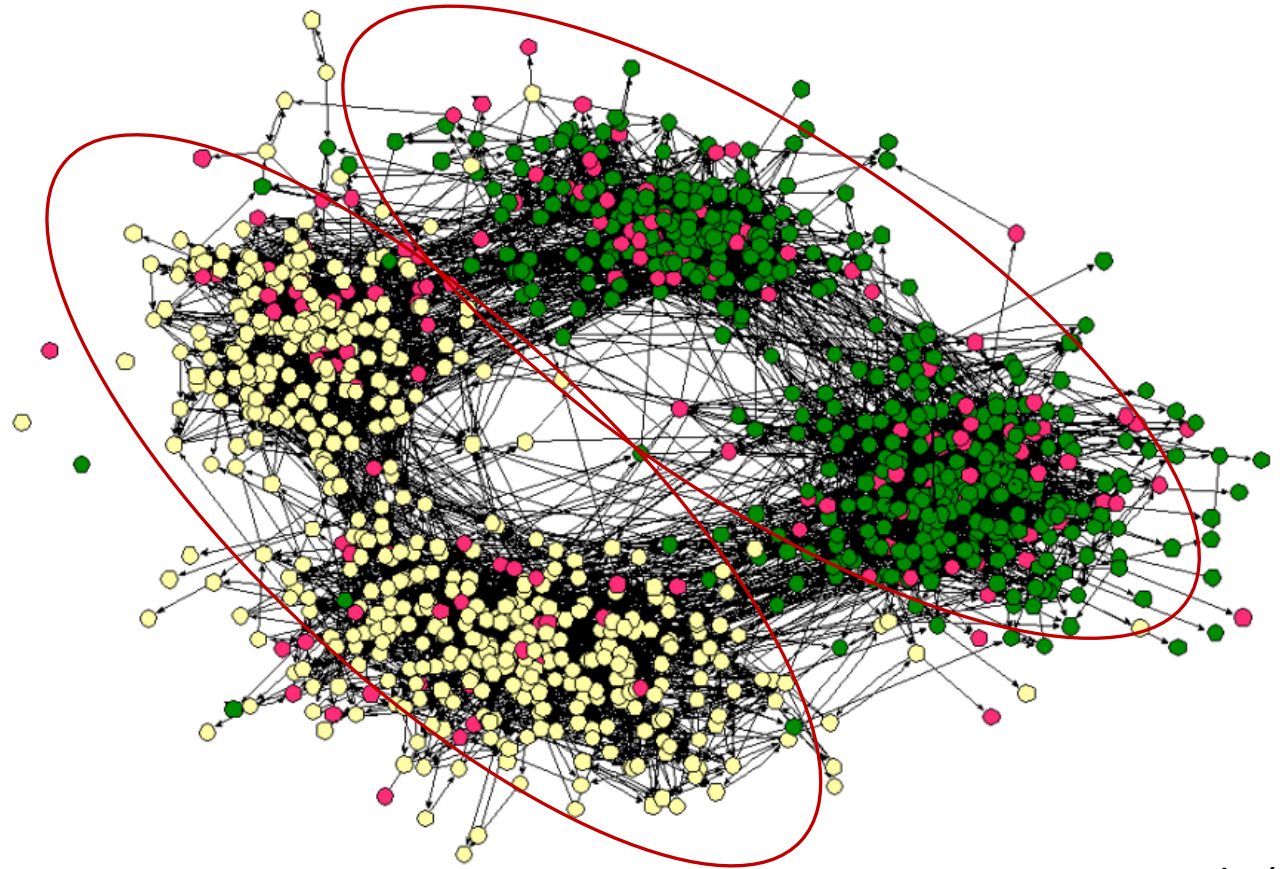


based on: *Networks, Crowds and Markets* by Easley & Kleinberg (2010)

Homophily, Social Influence, and Affiliation

A social network built by students from the town's middle and high schools

~ **Clustering by race**

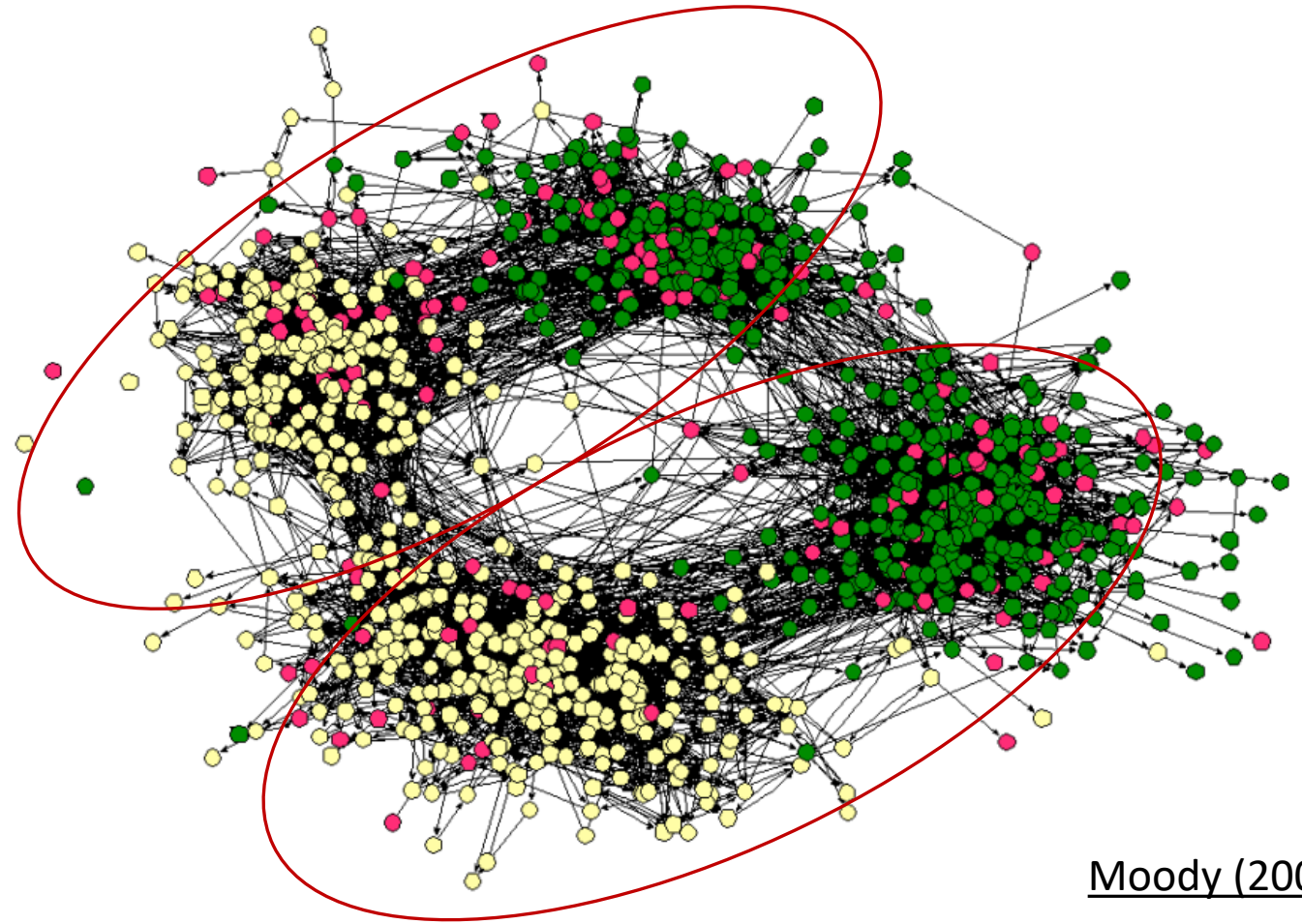


Moody (2001)

Homophily, Social Influence, and Affiliation

A social network built by students from the town's middle and high schools

~ **Clustering by age**



Moody (2001)

Homophily and Triadic Closure

Triadic closure:

- When two individuals share a common friend, a friendship between the two individuals is also more likely to occur
- Homophily suggests two individuals are more alike because of a common friend, so a link may occur even if neither is aware of a mutual friend!
- **Usually, it is difficult to attribute the formation of a link to any one factor**

==> Is there a simple test for the presence of homophily?

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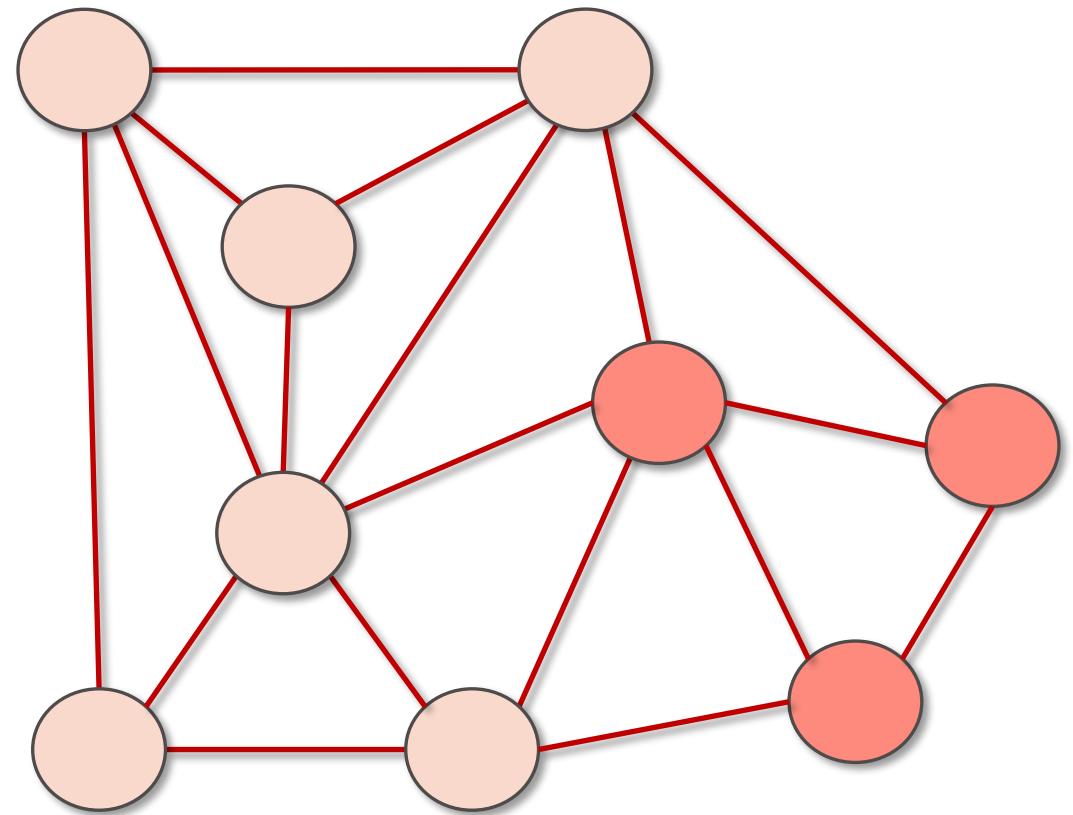
Homophily, Social Influence, and Affiliation

- **Homophily Test**
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- Affiliation Networks
- Schelling Model of Segregation
- Positive and Negative Relationships

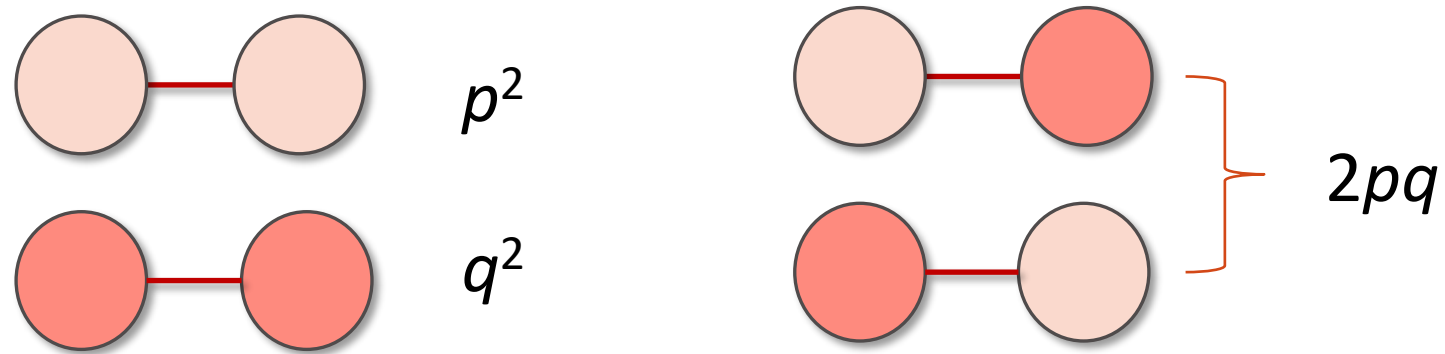
Homophily, Social Influence, and Affiliation

An example:
a male/female network

=> What would this network look like if it did *not* exhibit homophily?



Homophily, Social Influence, and Affiliation



A randomly generated network:

- If each node would be randomly assigned a gender according to the gender balance of the real network, the number of cross-gender edges should not change significantly relative to the real network
- If fraction p are males and q are females, what is the probability of...

Homophily, Social Influence, and Affiliation

A Homophily Test:

~ If the fraction of heterogeneous (cross-gender) edges is significantly less than $2pq$, then there is evidence for homophily

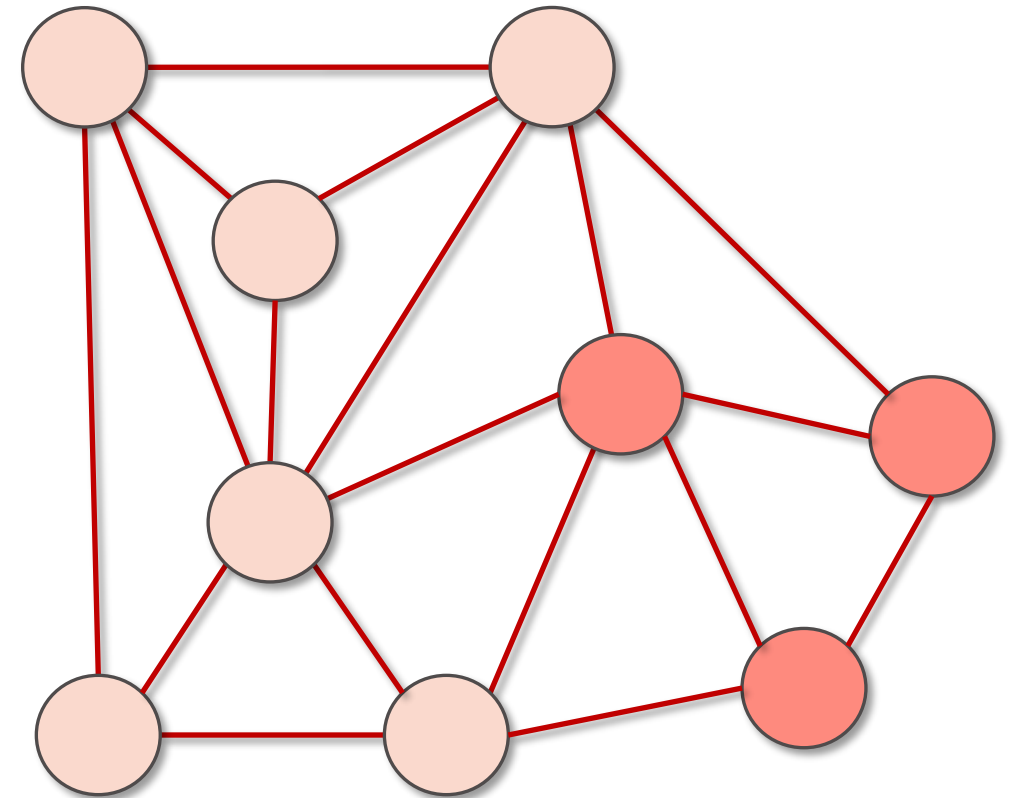
Cross-gender edges: 5 of 18

$$p = 6/9 = 2/3$$

$$q = 3/9 = 1/3$$

x without homophily, # of cross-gender edges should be $2pq = 4/9 = 8$ out of 18

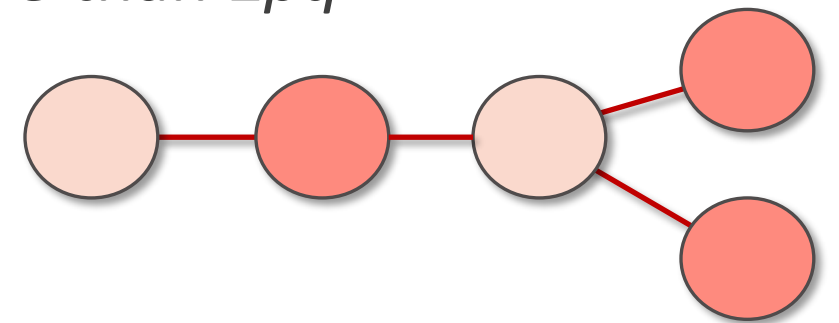
∴ Evidence of homophily



Homophily, Social Influence, and Affiliation

Notes on Homophily Test:

- “significantly less than” – How significant? A deviation below the mean is suitable
- What if the network had significantly *more* than $2pq$ cross-gender edges?
 - inverse homophily
 - **Example:** male-female dating relationships



Homophily, Social Influence, and Affiliation

Notes on the homophily test:

- Homophily test can be used to test any characteristic like race, age, native language, preferences, etc.
- For characteristics that have more than 2 values, perform the same type of calculation
- Compare the number of heterogeneous edges (edges with nodes that differ in interested characteristic) to what a randomly generated graph would look like using real data as probabilities

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Homophily, Social Influence, and Affiliation

Why is homophily often present in a social network?

➔ Selection

~ people tend to *choose* friends who are like themselves

- can operate at different scales and levels of intentionality
 - you actively choose friends that are like yourself among a small group of people
 - your school's population is relatively homogeneous compared to the overall population, so your environment compels you to choose friends like yourself

Homophily, Social Influence, and Affiliation

Mutable and Immutable Characteristics:

- Selection operates differently based on the type of characteristic
 - **Immutable:** Characteristics that don't change (gender, race) or change consistently with the population (age, generation)
 - **Mutable:** Characteristics that can change over time (behaviors, beliefs, interests, opinions)

Homophily, Social Influence, and Affiliation

Selection and Social Influence:

- Research has shown that people are susceptible to **social influence**
~ they may change their behaviors so as to more closely resemble the behaviors of their friends (surprise!)
- “Bad company corrupts good character.” – 1 Corinthians 15:33
- Social influence is the opposite of selection.
 - **Selection:** Individual characteristics drive the formation of links
 - **Social influence:** Existing links shape people’s mutable characteristics

Homophily, Social Influence, and Affiliation

Longitudinal Studies (long time span):

- Difficult to tell if selection or social influence is at play with a single snapshot of a network
- Longitudinal studies tracking social connections and individual behaviors over time can help researchers uncover the effect of social influence
- **Does behavior change after changes to network, or does network change after changes in behavior?**

Homophily, Social Influence, and Affiliation

Example:

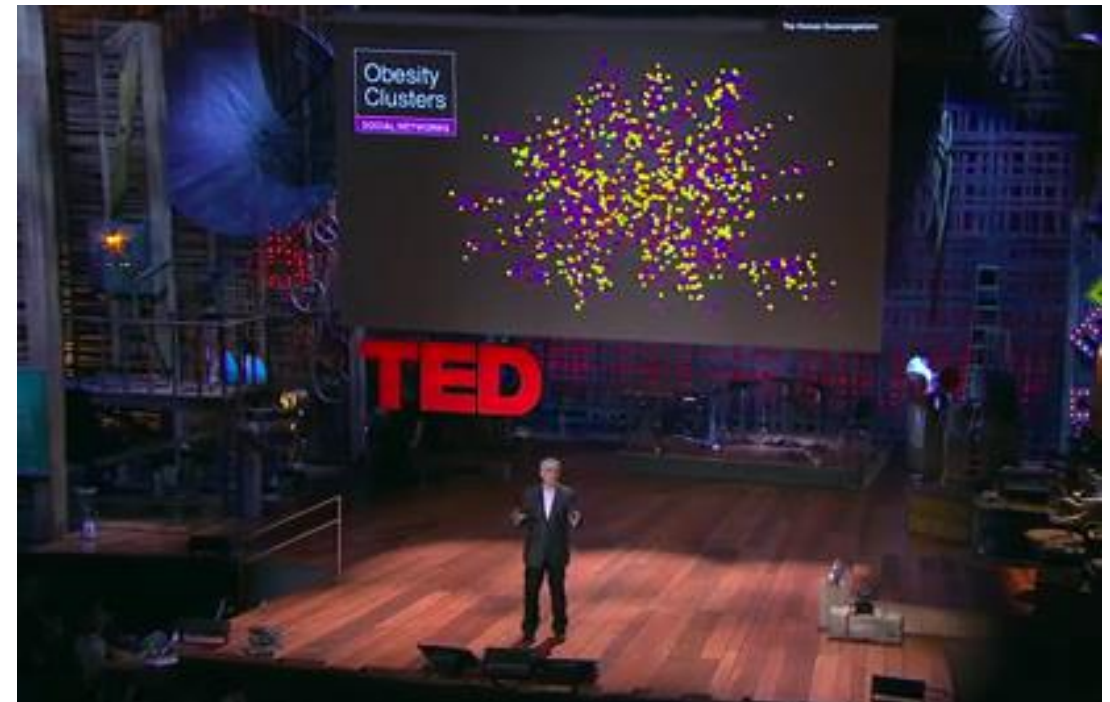
- Longitudinal studies have been used to determine if selection or social influence has had more an effect on teenagers' scholastic achievement and drug use
- Understanding these effects can be helpful in developing interventions
- If drug use displays homophily in a network, a program to target social influence (get friends to influence other friends to stop) might work best
- But if homophily took place due to selection, former drug users may choose new friends and the drug-using behavior of others is not strongly affected

Homophily, Social Influence, and Affiliation

Social Influence of Obesity:

- Christakis and Fowler tracked the obesity status and social network of 12,000 people over 32 years
- Homophily found based on obesity status

==> WHY?



Christakis & Fowler (2007), http://www.ted.com/talks/nicholas_christakis_the_hidden_influence_of_social_networks.html

Why does there occur obesity homophily?

- Selection effects:

Do people choose to befriend others of similar obesity status?

- Confounding effects of homophily:

Are there other factors, too, that correlate with obesity status?

- Social influence:

If friends changed their obesity status, did it influence a person's future obesity status?

➔ Discovered significant evidence for hypothesis 3:
Obesity is a type of “contagion” that can spread through social influence!

Social Networks and their Analysis

Influence Modeling

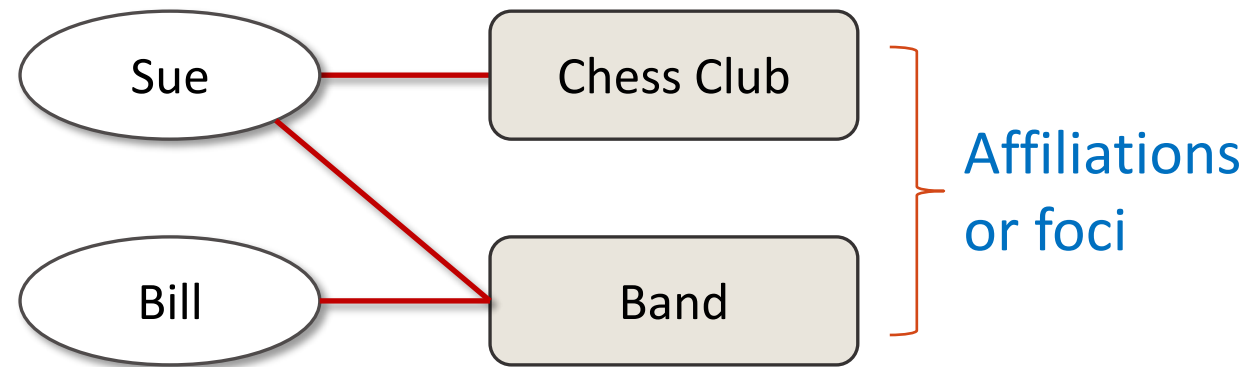
- *Linear Threshold Model*
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Homophily, Social Influence, and Affiliation

- Homophily Test
- Selection and Social Influence
- **Affiliation Networks**
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Affiliation Network

Putting context into the network by showing connections to activities, companies, organizations, neighborhoods, etc.

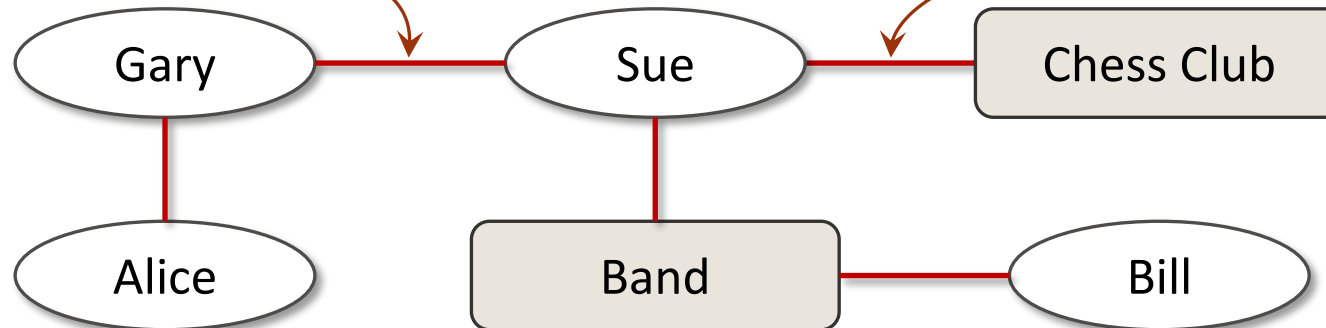


Bipartite graph: every edge joins two nodes belonging to different sets

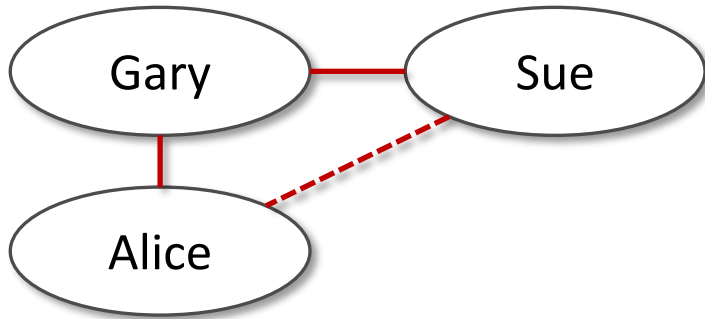
Social-Affiliation Network

Two types of edges:

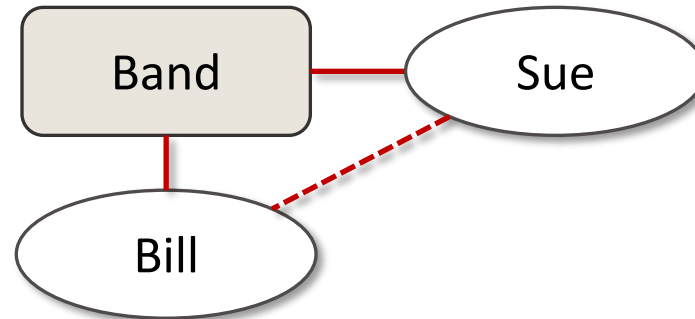
- Person to person: friendship or other social relationship
- Person to foci: participation in the focus



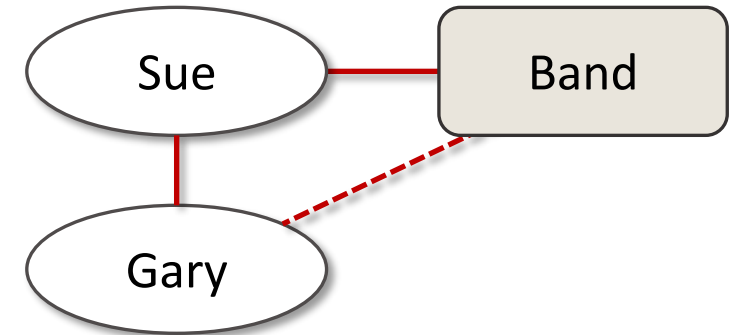
Closure Processes



Triadic closure



Focal closure: closure due to selection



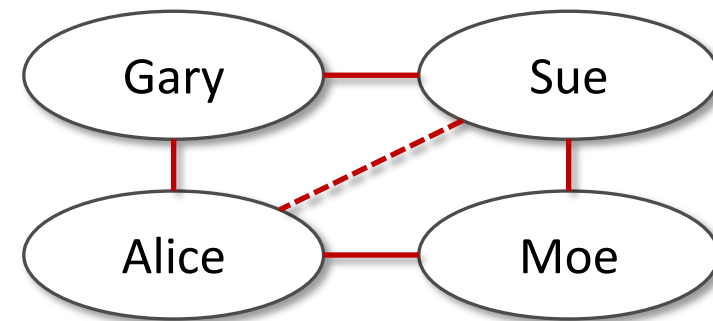
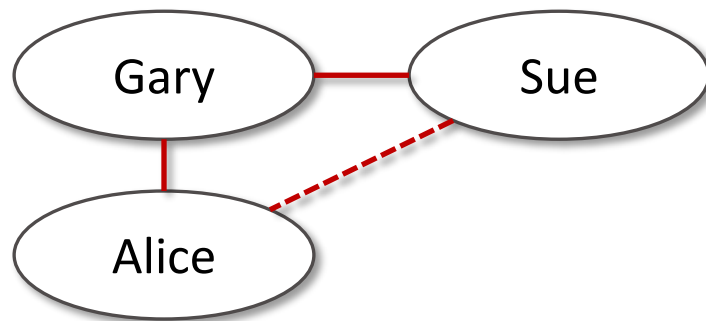
Membership closure: closure due to social influence

Questions About Closure

Would Sue be *more likely* to become friends with Alice if they shared more than one friend?

In other words, is triadic closure dependent on the number of shared friends?

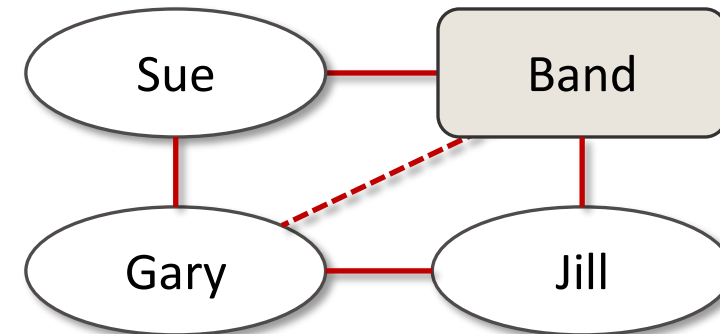
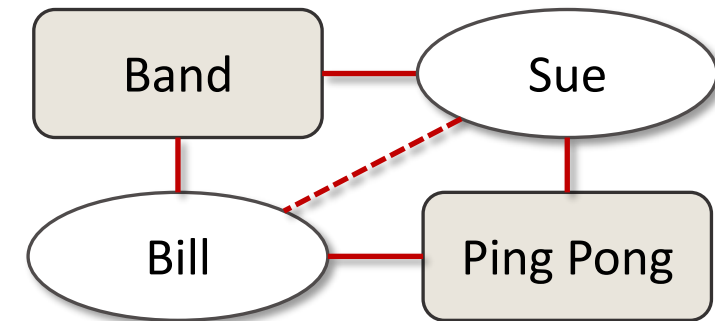
More formally: **What is the *probability* that two people form a link as a function of the number of mutual friends they share?**



Questions About Closure

For *focal closure*, what is the probability that two people form a link as a function of the number of foci they are jointly affiliated with?

For *membership closure*, what is the probability that a person becomes involved in a particular focus as a function of the number of friends who are already involved in it?



Methodology: Measuring Triadic Closure

1. Take two snapshots of the network at times t_1 and t_2
2. For each k , identify all pairs of nodes who have exactly k friends in common at t_1 but who are not directly connected by an edge
3. Define $T(k)$ to be the fraction of these pairs that form an edge by t_2 . This is the probability that a link will form between two people with k friends in common
4. Plot $T(k)$ as a function of k to illustrate the effect of common friends on link formation

Another example: Email Social Network

Kossinets and Watts (2006) examined an email communication of 22,000 students over one year at a large US university

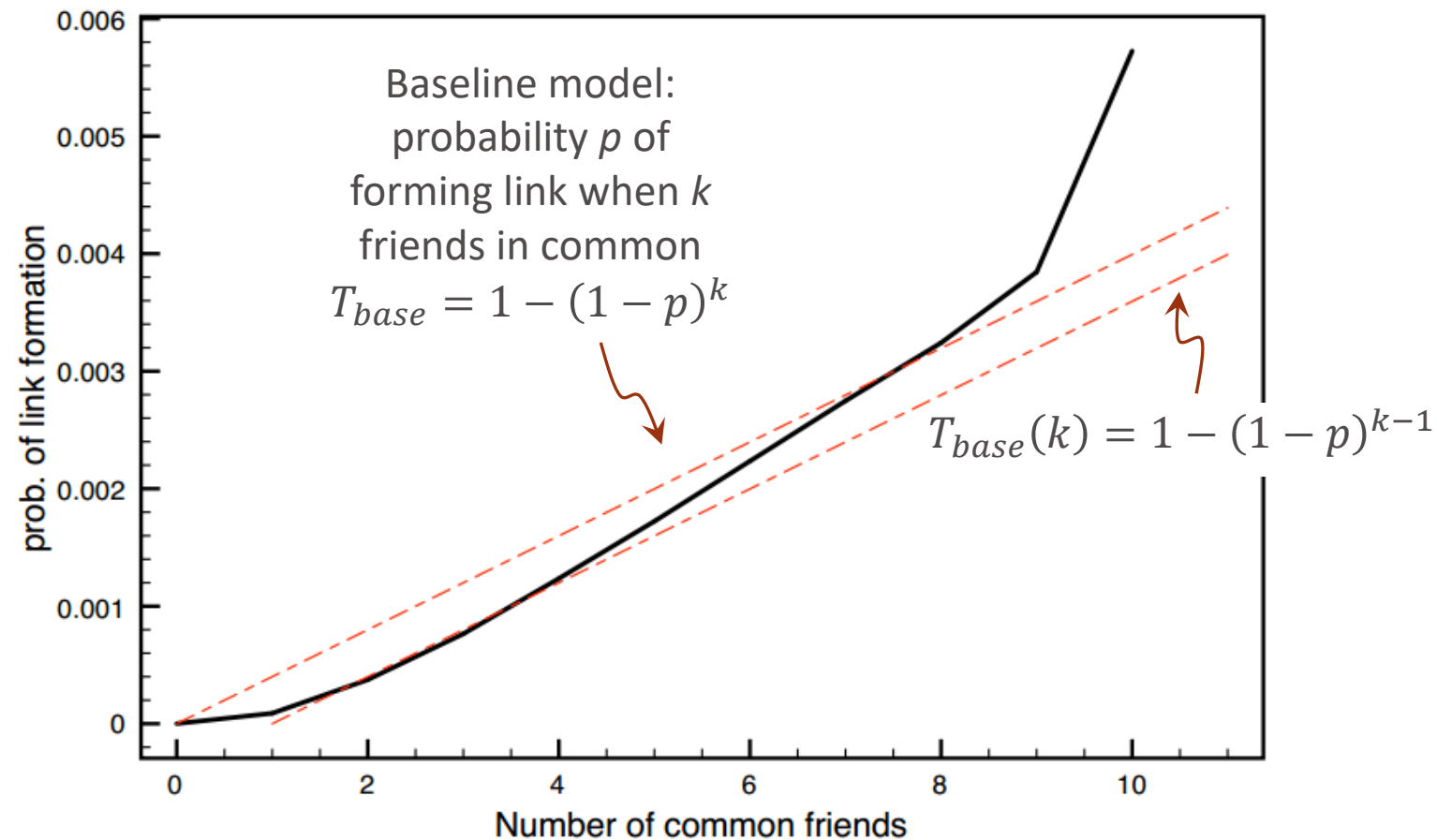
Assumed a link between two people if an email was sent between the two in the past 60 days

Each snapshot is one day apart

$T(k)$ averaged over multiple pairs of snapshots

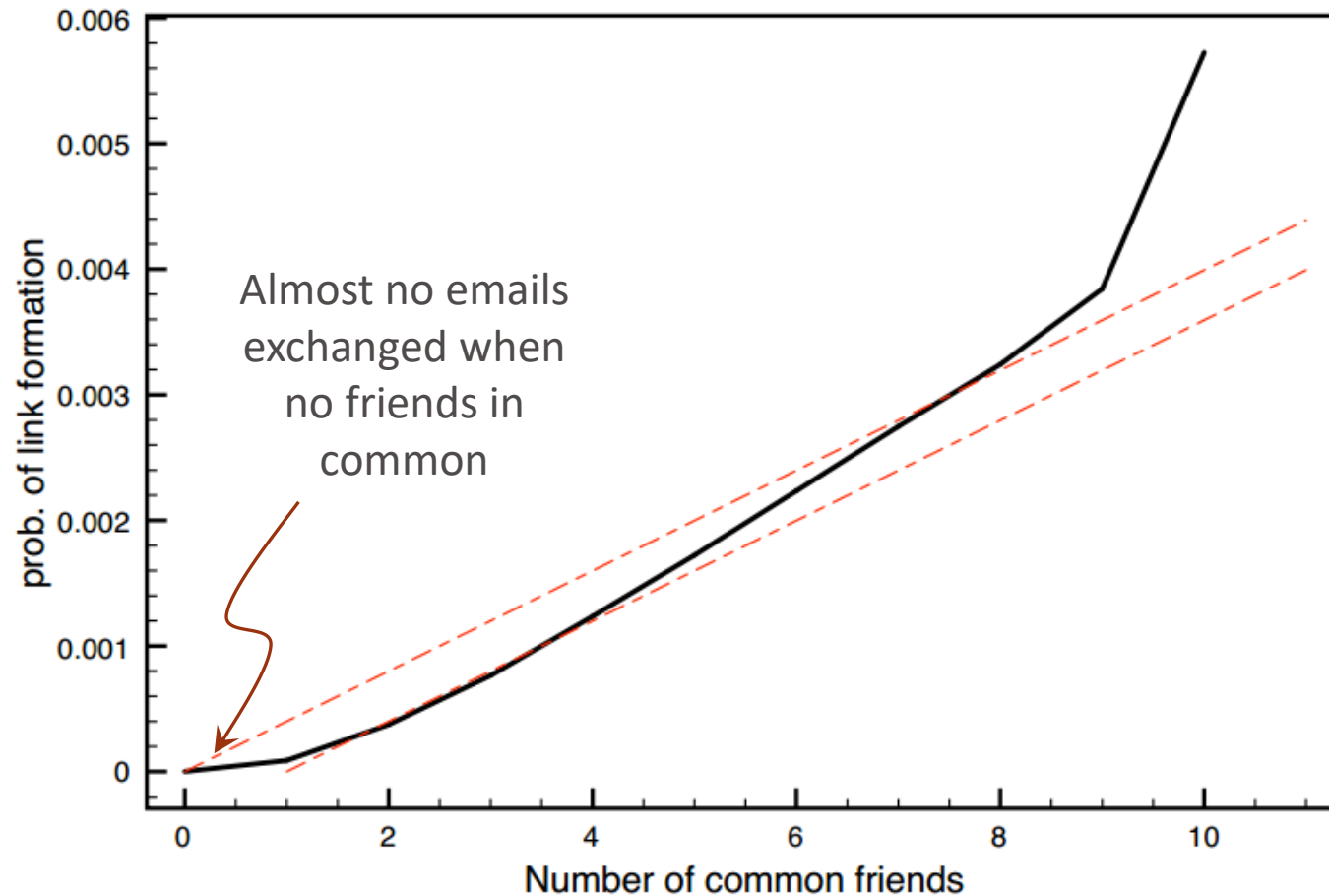
Triadic Closure in the Email Data Set

Figure from <http://www.cs.cornell.edu/home/kleinber/networks-book/networks-book-ch04.pdf>



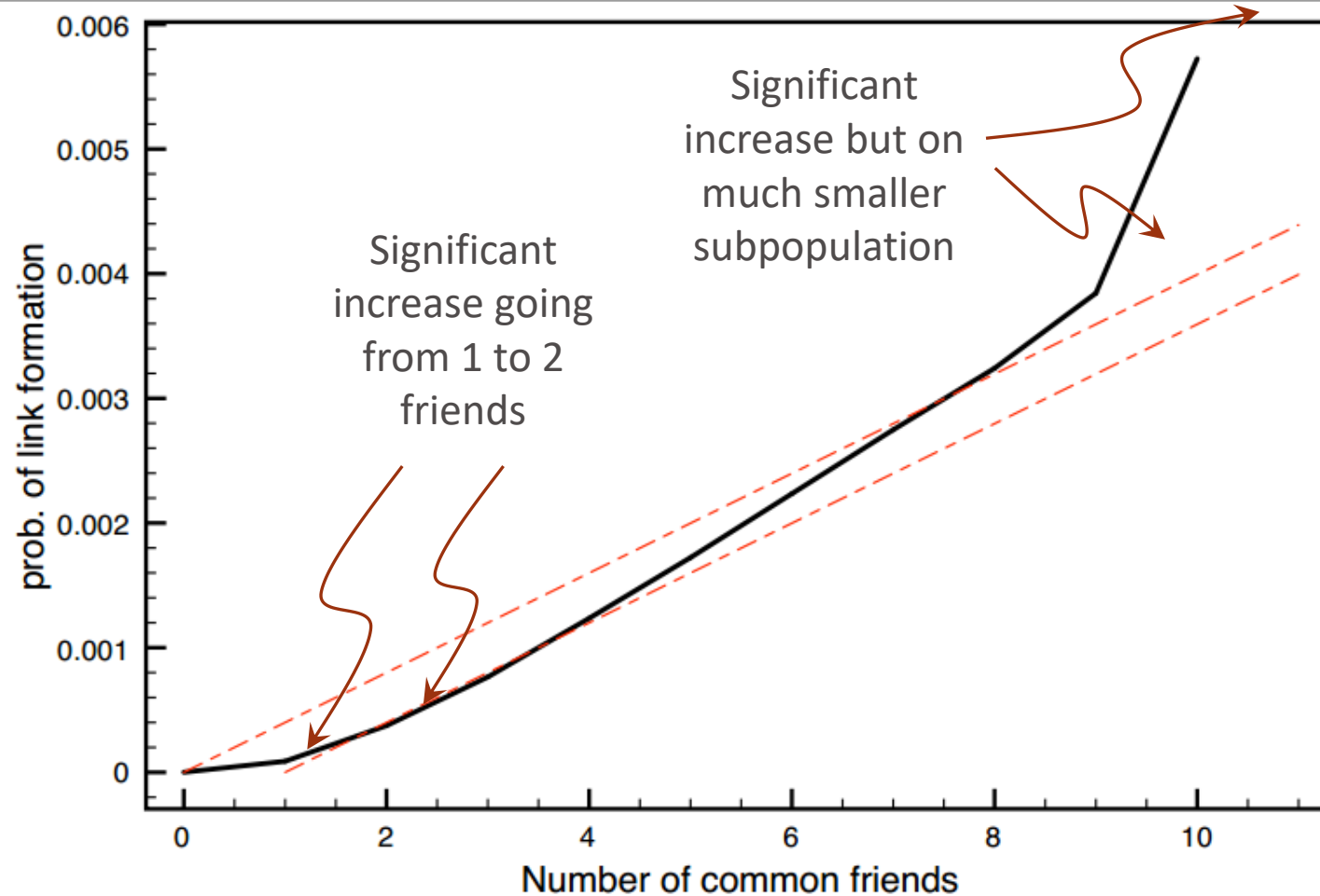
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Triadic Closure in the Email Data Set

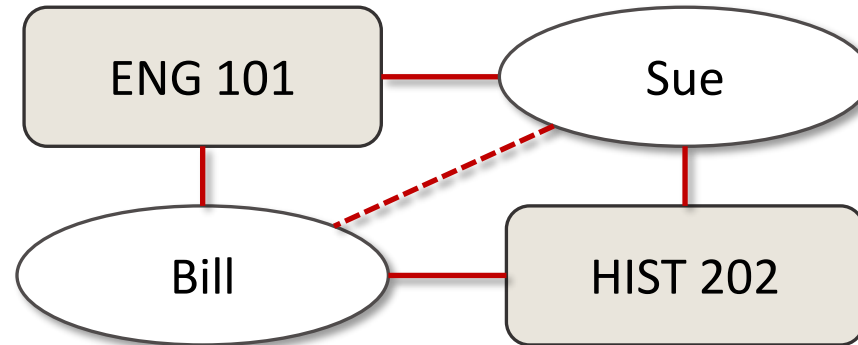
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Email Social Network

To evaluate focal closure, Kossinets and Watts obtained class schedules for 22,000 students

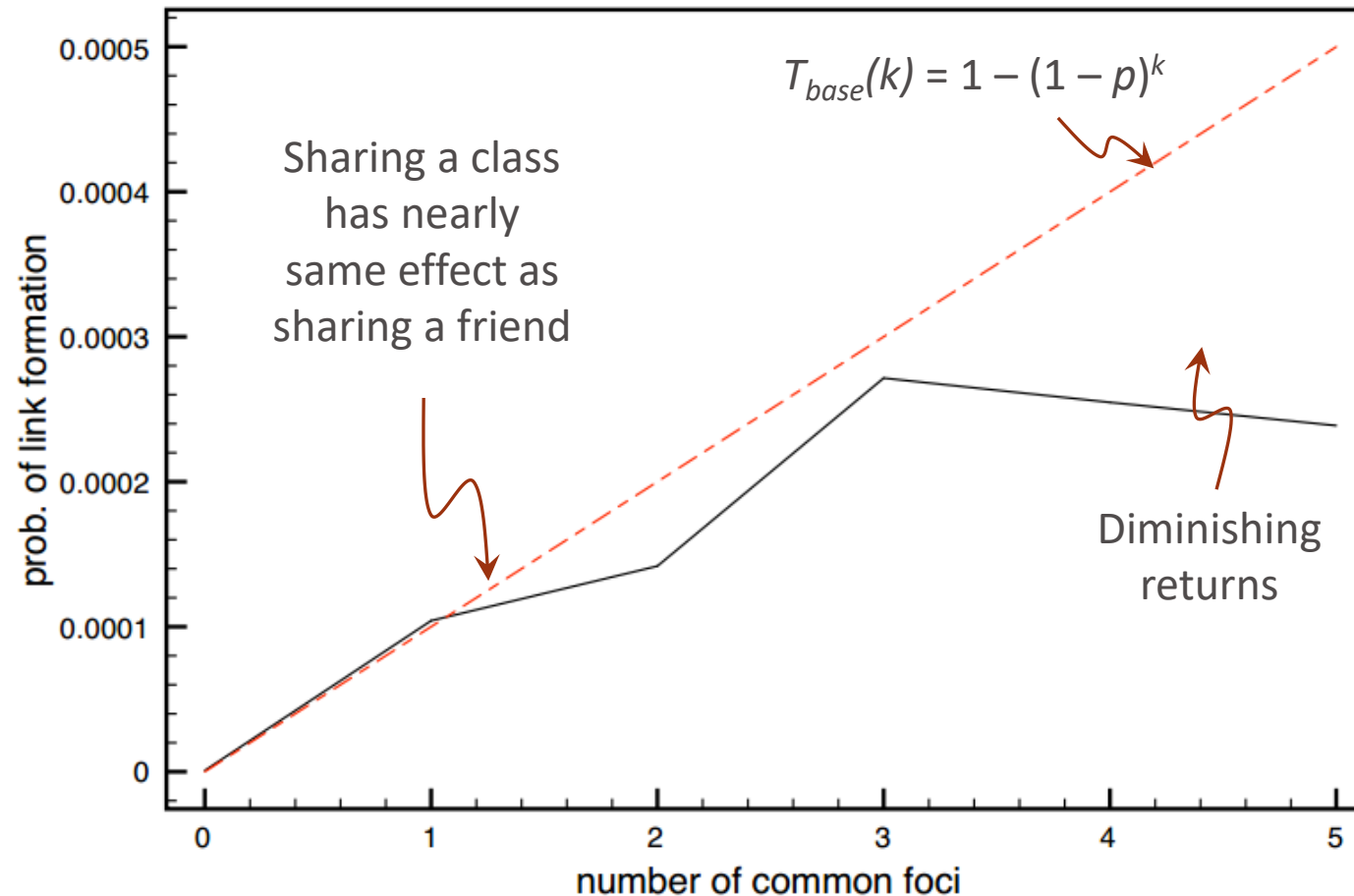
Created a social-affiliation network where classes are foci



Determined probability of link formation as a function of the number of shared foci

Focal Closure in the Email Data Set

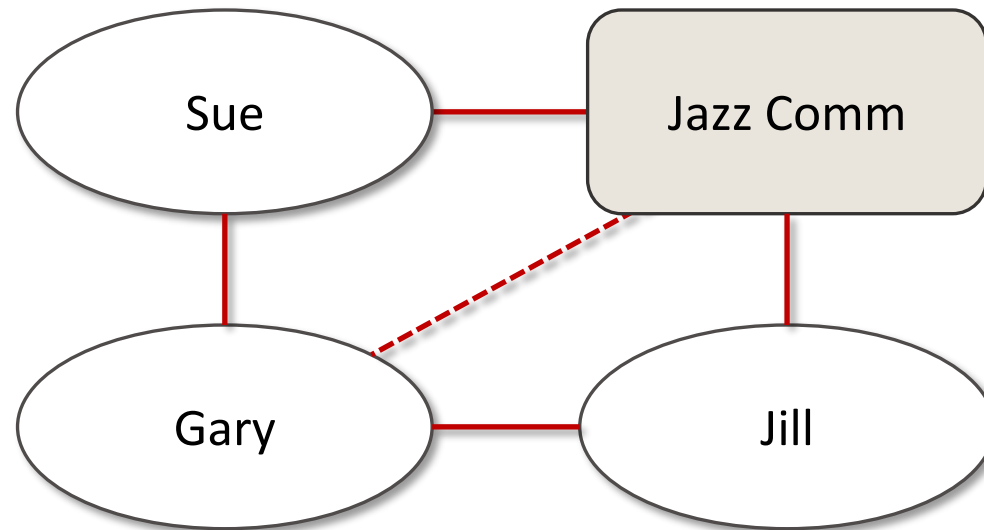
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Measuring Membership Closure

Backstrom et al. (2006) created a social-affiliation network for LiveJournal

- Friendships designated by users in their profile
- Foci are membership in user-defined communities



Membership Closure in LiveJournal Data Set

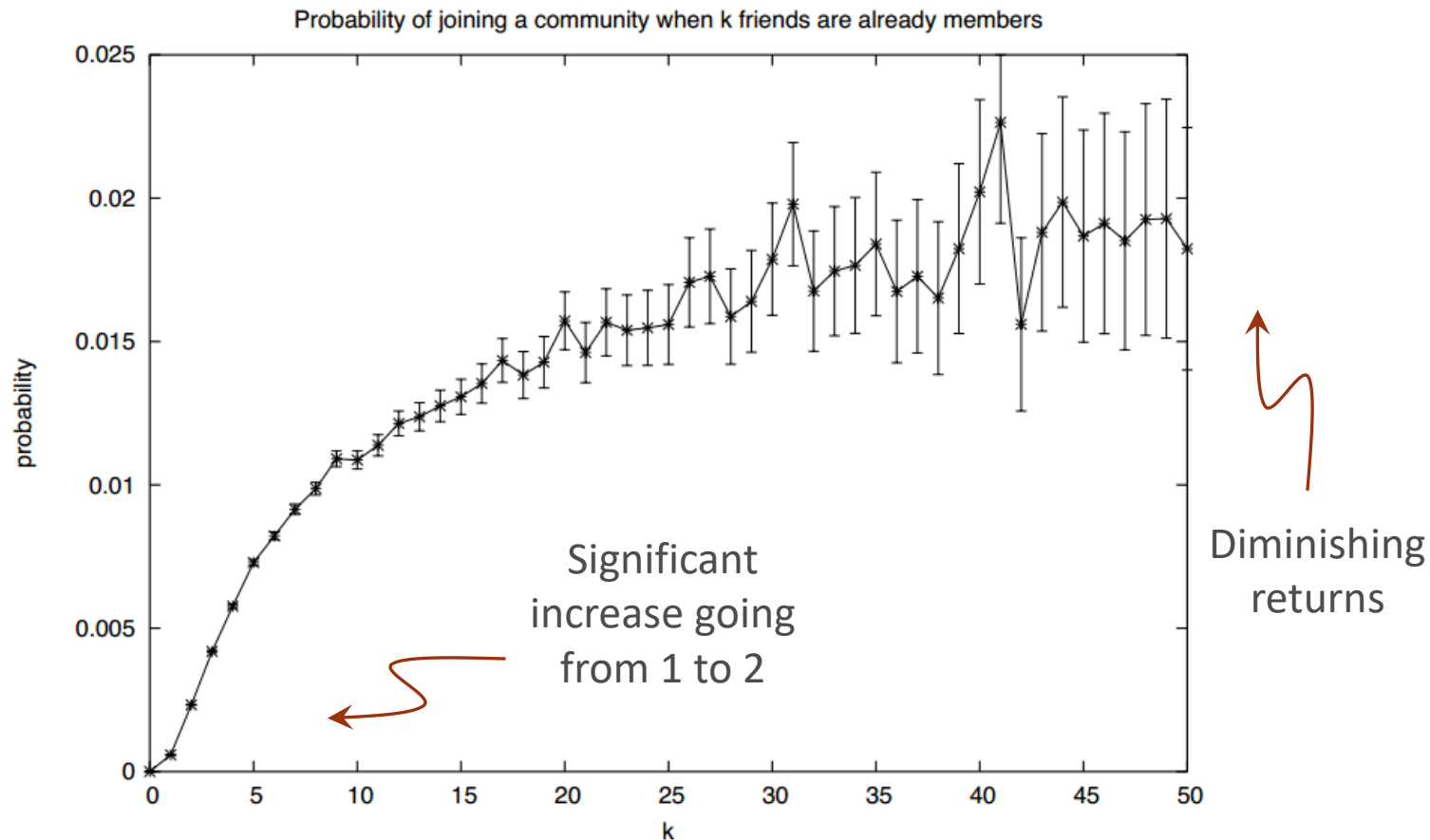
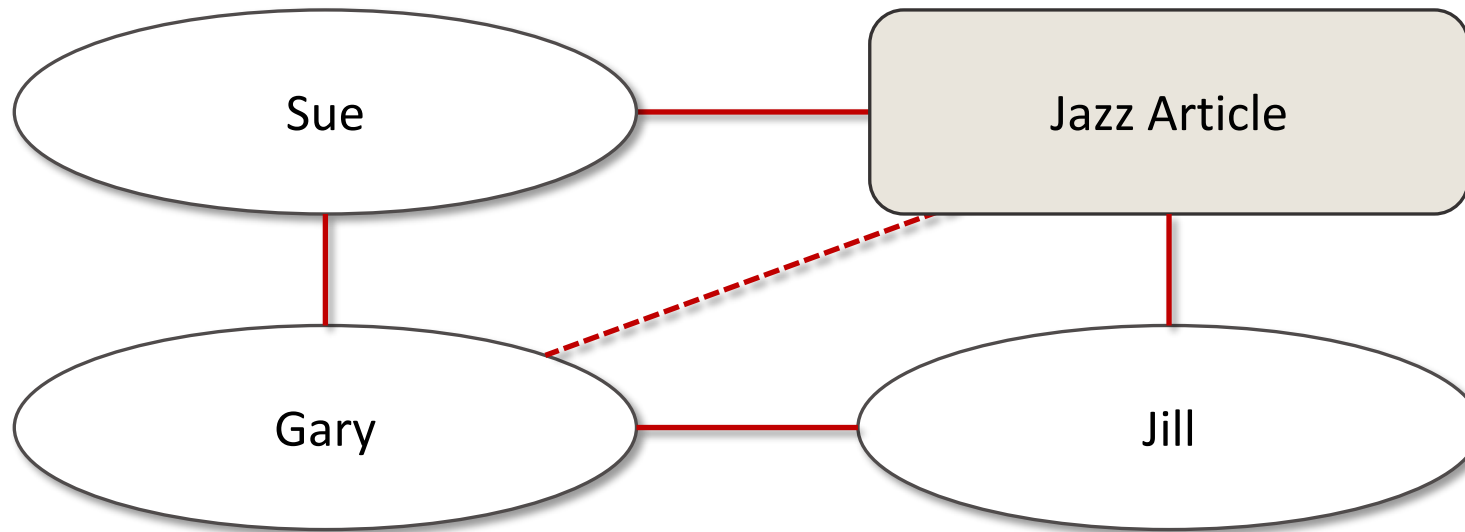


Figure from <http://www.cs.cornell.edu/home/kleinber/networks-book/networks-book-ch04.pdf>

Measuring Membership Closure in the Wikipedia Data Set

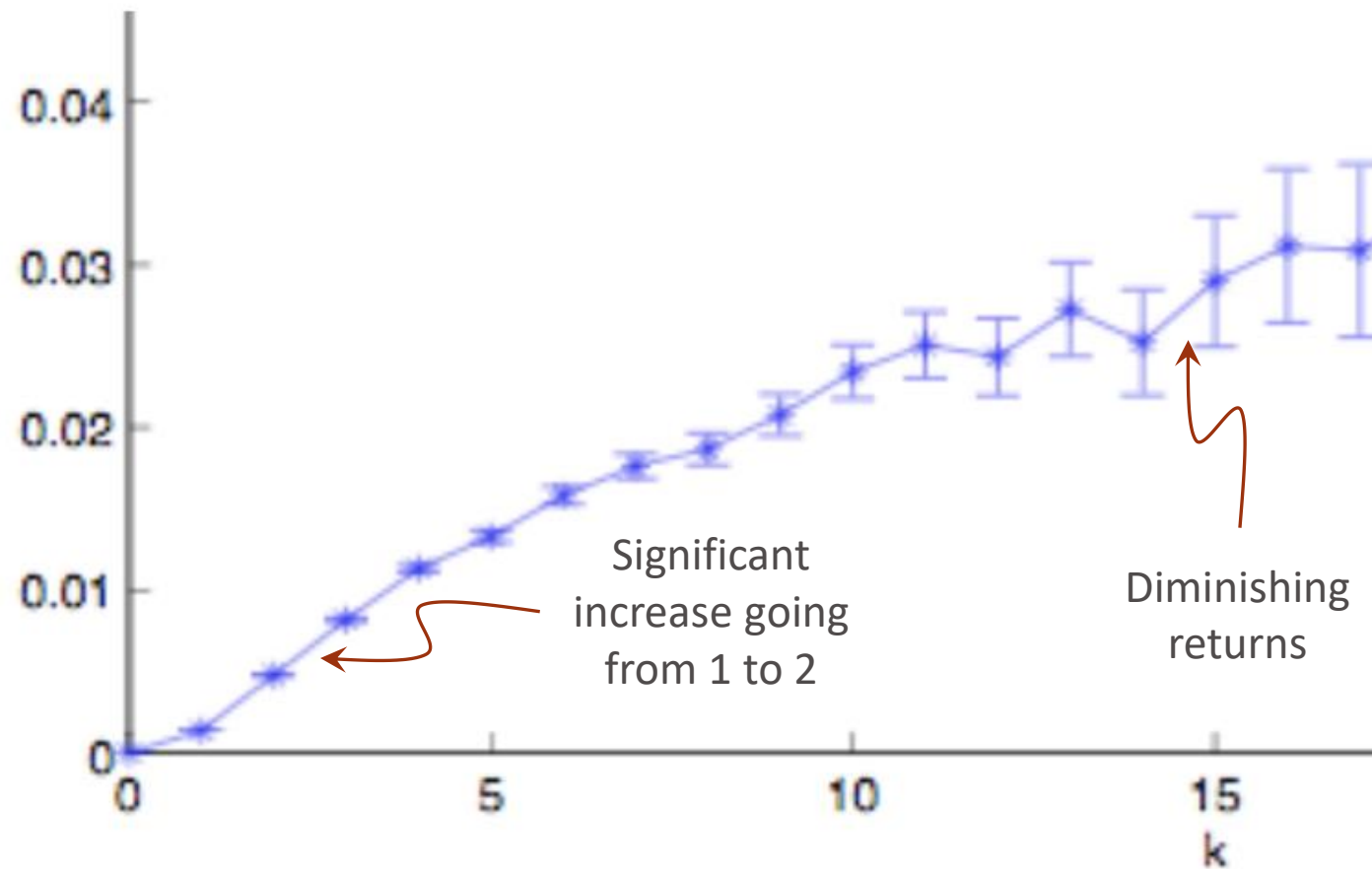
Crandall et al. (2008) created a social-affiliation network for Wikipedia

- Node for editors maintaining a user talk page
- Link if two editors have communicated using talk page
- Foci are articles edited by editors



Membership Closure in the Wikipedia Data Set

Figure from <http://www.cs.cornell.edu/home/kleinber/networks-book/networks-book-ch04.pdf>



Selection and Social Influence in the Wikipedia Data Set

A further examination of the Wikipedia data set to see evidence of homophily produced by selection and social influence

How to measure the similarity of two editors?

$$\frac{\text{number of articles edited by both editors}}{\text{number of articles edited by at least one editor}}$$

Evaluated for editors in an affiliation network of editors and articles
Similar to the *neighborhood overlap* for edges

Selection and Social Influence in the Wikipedia Data Set

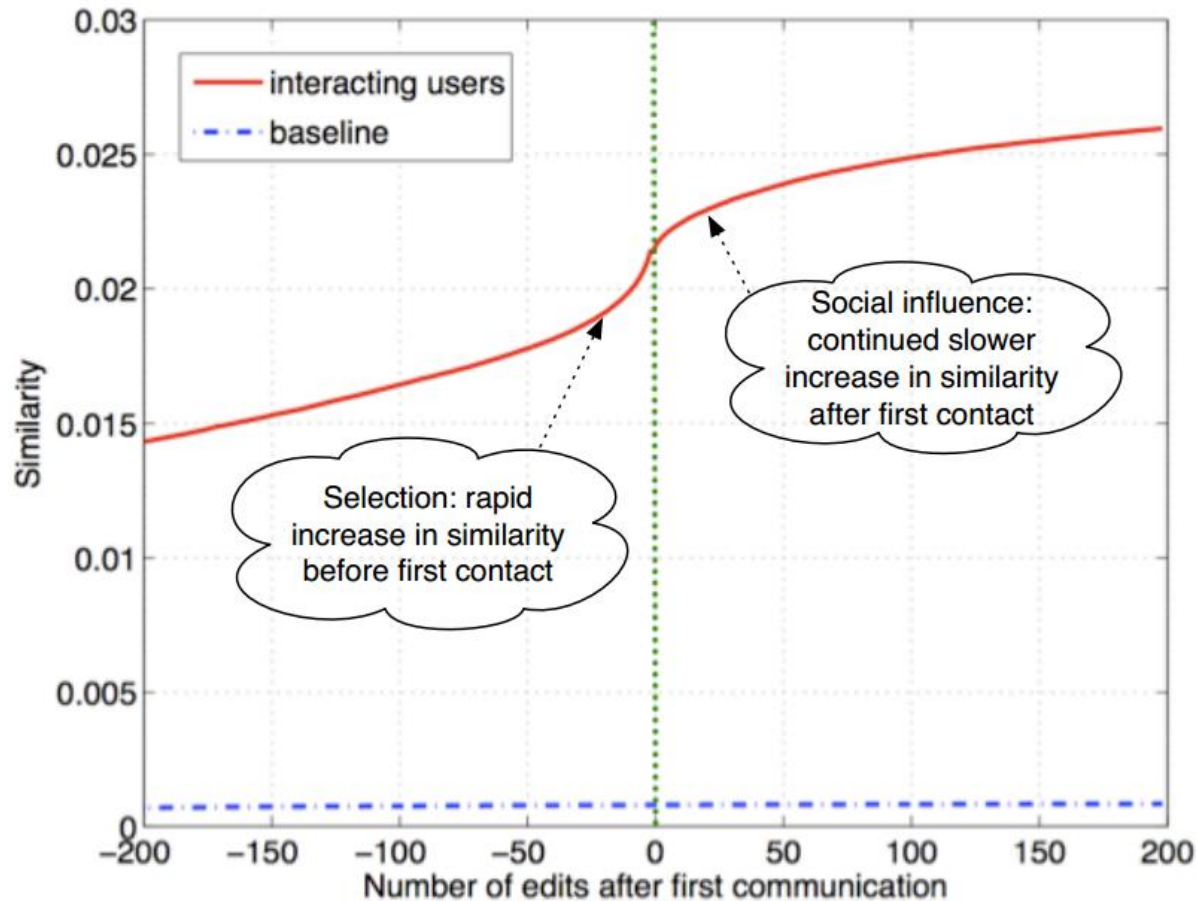
Pairs of Wikipedia editors who have communicated are significantly more similar in behavior than editors who have never communicated

Homophily arises because...

- editors are talking with editors who have edited the same article? (selection)
- editors are led to the articles by those they talk to? (social influence)

Selection and Social Influence in the Wikipedia Data Set

Figure from <http://www.cs.cornell.edu/home/kleinber/networks-book/networks-book-ch04.pdf>



Average of interaction histories

- 0 corresponds to time when the link occurred (friendship – they talk to each other)
- Similarity measured relative to time when they became friends
- Only editors that have at least 100 actions both before and after the first interaction.
- Baseline: the similarity of random pair of Wikipedia editors

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Homophily, Social Influence, and Affiliation

Segregation:

~ Cities are often divided into homogeneous neighborhoods based on ethnic and racial lines



Image: <http://wideurbanworld.blogspot.com/2011/08/race-ethnicity-social-class-are-most.html>

Homophily, Social Influence, and Affiliation

Racial distribution in **St. Louis**, 2010 U.S. Census. Each dot is 25 people: ● White ● Black ● Asian ● Hispanic ● Other

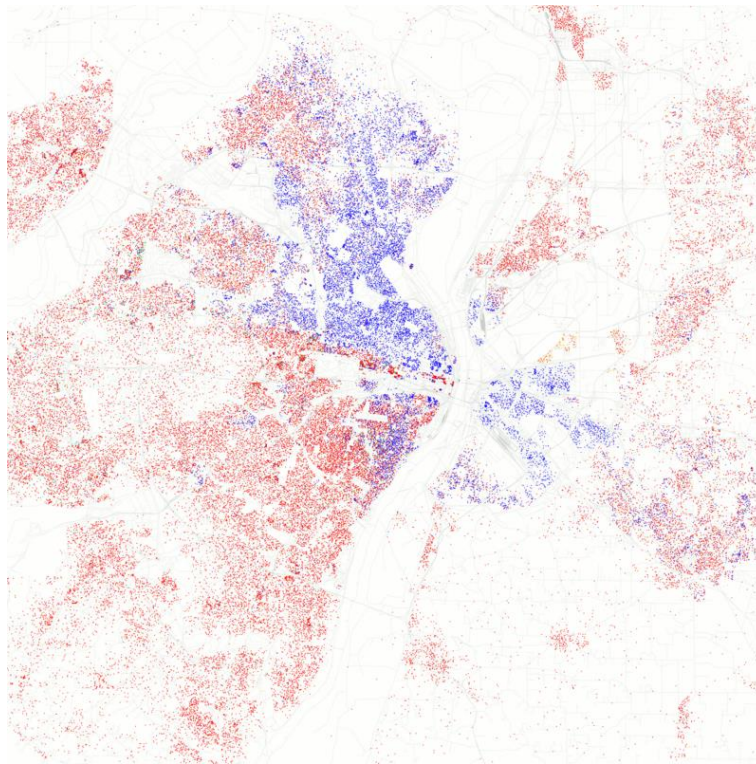
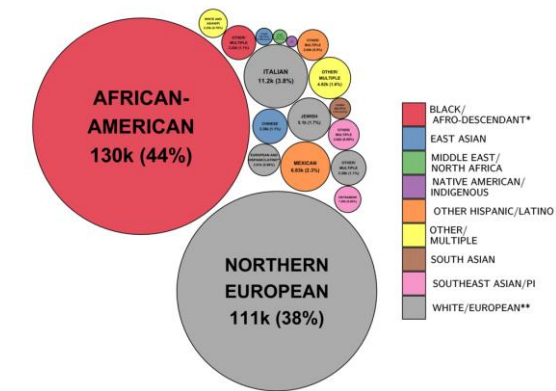


Image: <http://wideurbanworld.blogspot.com/2011/08/race-ethnicity-social-class-are-most.html>

Ethnic Origins: St. Louis in 2021
Total population: 294,248



Segregation:

~ Cities are often divided into homogeneous neighborhoods based on ethnic and racial lines

- Desegregation efforts (e.g., forced busing), „white fly,” and suburbanization

Homophily, Social Influence, and Affiliation

Segregation:

~ Cities are often divided into homogeneous neighborhoods based on ethnic and racial lines

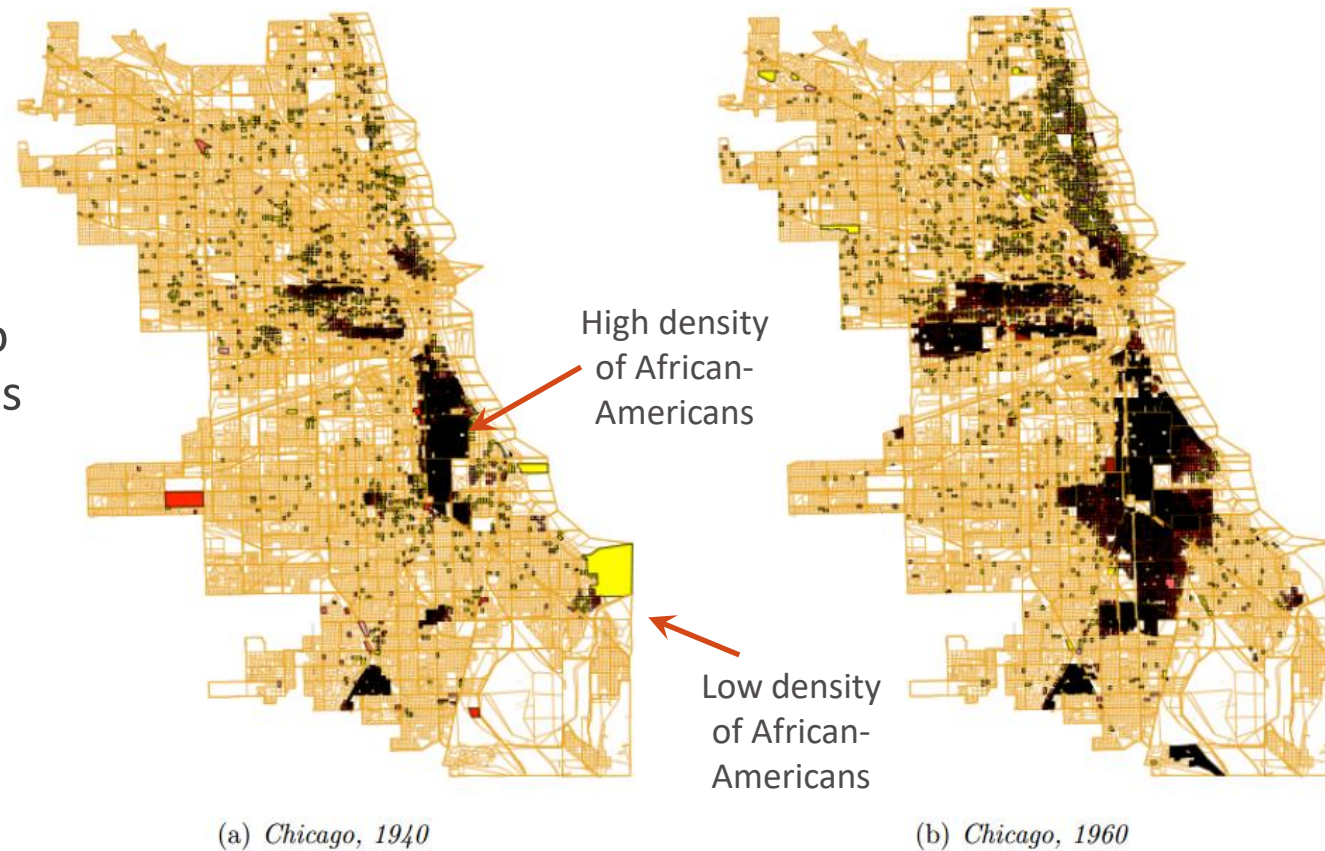


Figure from <http://www.cs.cornell.edu/home/kleinber/networks-book/networks-book-ch04.pdf>

Homophily, Social Influence, and Affiliation

Schelling Model of Segregation:

- a simple model introduced by Thomas Schelling in the 1970s
- **shows how global patterns of self-chosen segregation might occur because of homophily at the local level**
- the population is formed by two types of individuals
 - **agents** (X and O in our case)
- two types of agents represent **immutable characteristics**
 - race, ethnicity, country of origin, or native language

Homophily, Social Influence, and Affiliation

Schelling Model of Segregation:

Agents placed
randomly in grid



Agents are **satisfied** with their location in the grid if at least t of their neighbors are agents of the same type.

t denotes a (pre-set) threshold

X	X	O		
X	O	O	O	
X	X		X	X
X	O		X	X
	O	O		

Homophily, Social Influence, and Affiliation

Schelling Model of Segregation:

- In each round, first identify all unsatisfied agents
- Assume t is 3 – all agents prefer to have at least 3 neighboring agents that are of the same type

X	X	O		
X	O	O	O	
X	X		X	X
X	O		X	X
	O	O		

Homophily, Social Influence, and Affiliation

Schelling Model of Segregation:

Unsatisfied because only
two neighbors are X



Assume t is 3 – all agents prefer to
have at least 3 neighboring agents that
are of the same type

X	X	O		
X	O	O	O	
X	X		X	X
X	O		X	X
	O	O		

Homophily, Social Influence, and Affiliation

Schelling Model of Segregation:

Unsatisfied because only
two neighbors are X

Assume t is 3 – all agents prefer to
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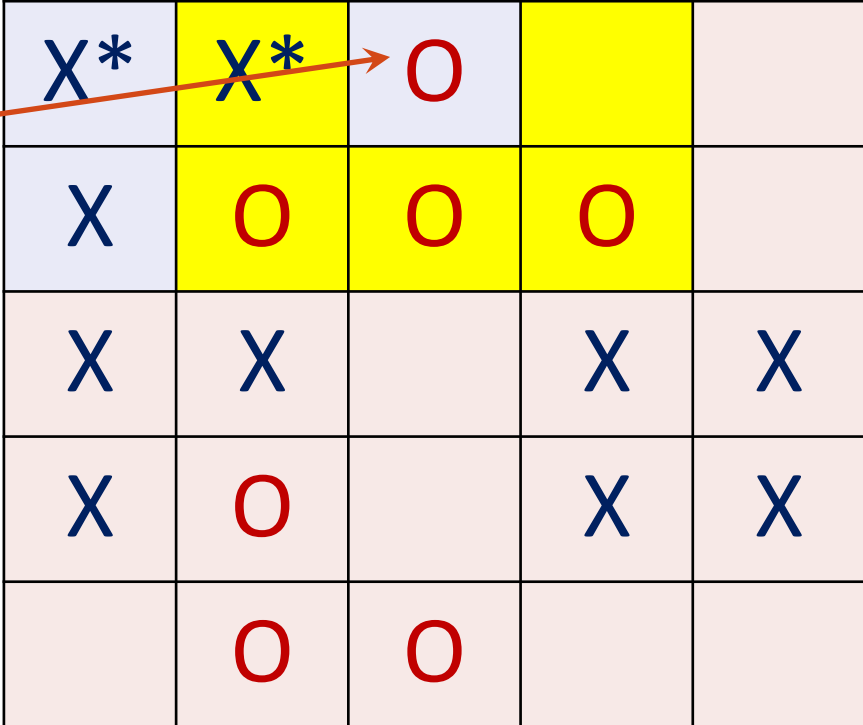
X*	X	O		
X	O	O	O	
X	X		X	X
X	O		X	X
	O	O		

Homophily, Social Influence, and Affiliation

Schelling Model of Segregation:

Satisfied because 3
neighbors are O

Assume t is 3 – all agents prefer to
have at least 3 neighboring agents that
are of the same type



X*	X*	O		
X	O	O	O	
X	X		X	X
X	O		X	X
	O	O		

Homophily, Social Influence, and Affiliation

Schelling Model of Segregation:

Satisfied because 4
neighbors are X



X*	X*	O		
X	O	O	O	
X	X		X	X
X	O		X	X
	O	O		

Assume t is 3 – all agents prefer to have at least 3 neighboring agents that are of the same type

Homophily, Social Influence, and Affiliation

Schelling Model of Segregation:

Unsatisfied

Assume t is 3 – all agents prefer to have at least 3 neighboring agents that are of the same type

X*	X*	O		
X	O	O	O	
X	X		X	X
X	O		X	X
	O	O		

Homophily, Social Influence, and Affiliation

Schelling Model of Segregation:

Assume t is 3 – all agents prefer to have at least 3 neighboring agents that are of the same type

Satisfied

X*	X*	O		
X	O*	O	O	
X	X		X	X
X	O		X	X
	O	O		

Homophily, Social Influence, and Affiliation

Schelling Model of Segregation:

Assume t is 3 – all agents prefer to have at least 3 neighboring agents that are of the same type

Unsatisfied

X*	X*	O		
X	O*	O	O	
X	X		X	X
X	O		X	X
	O	O		

Homophily, Social Influence, and Affiliation

Schelling Model of Segregation:

- etc.
 - ➔ the final result where unsatisfied agents marked with *
- Now, all unsatisfied agents move to a location where they are satisfied
- Assumed t was 3 – all agents prefer to have at least 3 neighboring agents that are of the same type

X*	X*	O		
X	O*	O	O*	
X	X		X	X
X*	O*		X	X
	O*	O*		

Homophily, Social Influence, and Affiliation

Schelling Model of Segregation:

=> The grid is now more segregated than before!

Previously satisfied agent
no longer satisfied!

Sometimes an agent can't find a new
location that will satisfy – leave alone
or move to random location

		O	O	O
X		O	O	O
X	X	X	X	X
	X	X	X	X
		O*		

Homophily, Social Influence, and Affiliation

Schelling Model of Segregation – Model Variations:

- Schedule agents to move in random order or in sweeping motion
- Agents move to the nearest location that satisfies or a random one
- Threshold could be percentage or vary among or within population groups
- World could wrap (left edge meets up with right edge)
- Many other variations, but *results usually end up the same:*

=> self-imposed segregation!

Homophily, Social Influence, and Affiliation

Schelling Model of Segregation – Simulators:

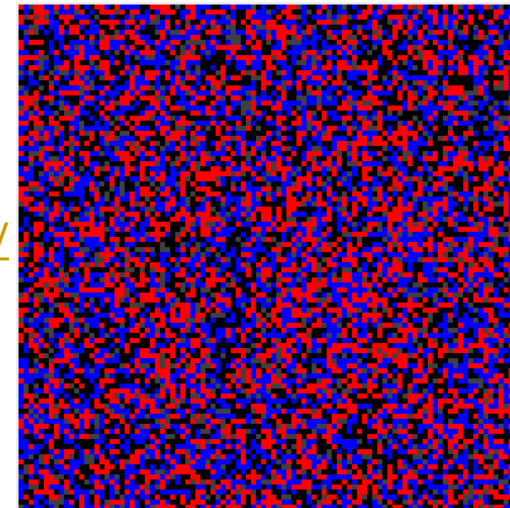
Uri Wilensky's NetLogo Segregation model

<http://ccl.northwestern.edu/netlogo/models/Segregation>

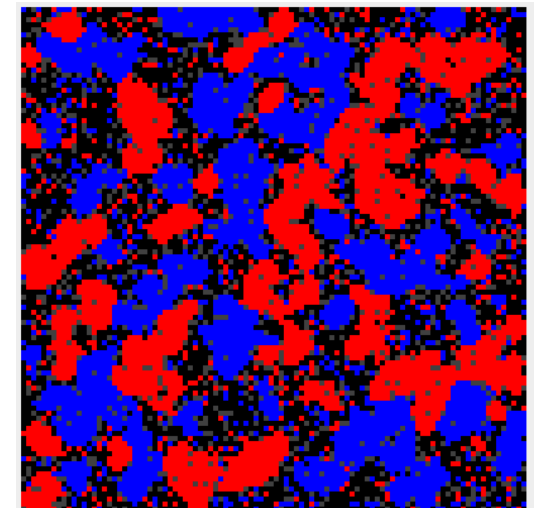
Sean Luke

<http://cs.gmu.edu/~eclab/projects/mason/projects/schelling/>

Luke Sean's Schelling Model using $t = 3$



Randomly placed



After 20 rounds

Homophily, Social Influence, and Affiliation

Schelling Model of Segregation – Conclusions:

- Schelling model is **an example of how immutable characteristics can become highly correlated with mutable characteristics**
- **Choice of where to live (mutable) over time conforms to agents' type (immutable), producing segregation (homophily)**

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- **Positive and Negative Relationships**

Positive and Negative Relationships

Till now, mostly a relation in a network has a positive connotation, but there exist social networks where also negative relationships can be expressed – antagonistic (enemy).

In the following part

- Positive relations – friendly
- Negative relations – enemy

Question: when is the network in a balanced state?

- We will see how local properties enforce global properties, cf. Schelling Model of Segregation

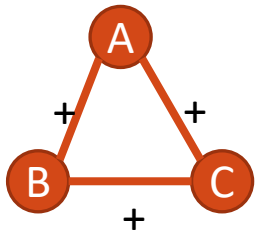
Note: we will greatly oversimplify – e.g., we consider networks where **all pairs of nodes are connected** by a positive or negative relation. Later we will relax it.

Positive and Negative Relationships

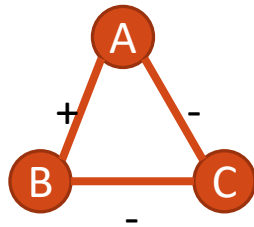
All nodes are connected = complete graph = clique

Applicable to small groups of nodes (people) or countries

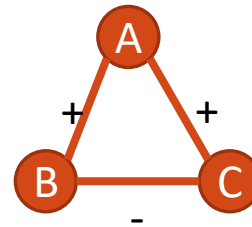
Four (up to) possible configurations of positive/negative edges among three nodes



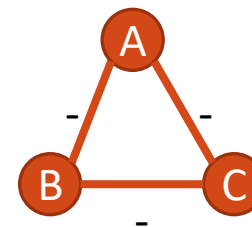
A, B, and C are mutual friends: **balanced**.



A and B are friends, with C as a mutual enemy: **balanced**.

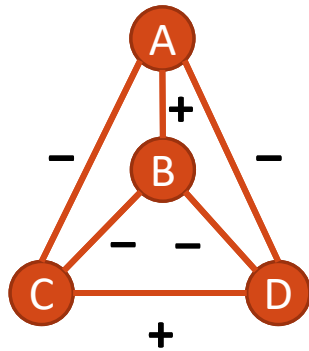


B and C are friends of A, but they don't get along with each other: **not balanced**.

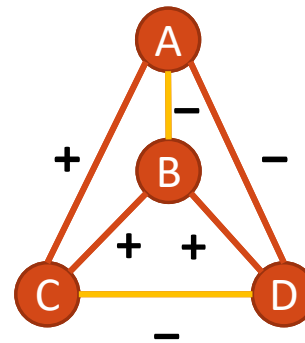


A, B, and C are mutual enemies: **not balanced**.

Positive and Negative Relationships



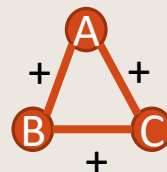
balanced



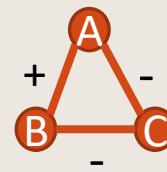
not balanced

Structural balance property:

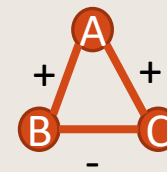
For every triangle edges, all three are positive or exactly one is positive.



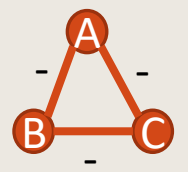
balanced



balanced



not balanced

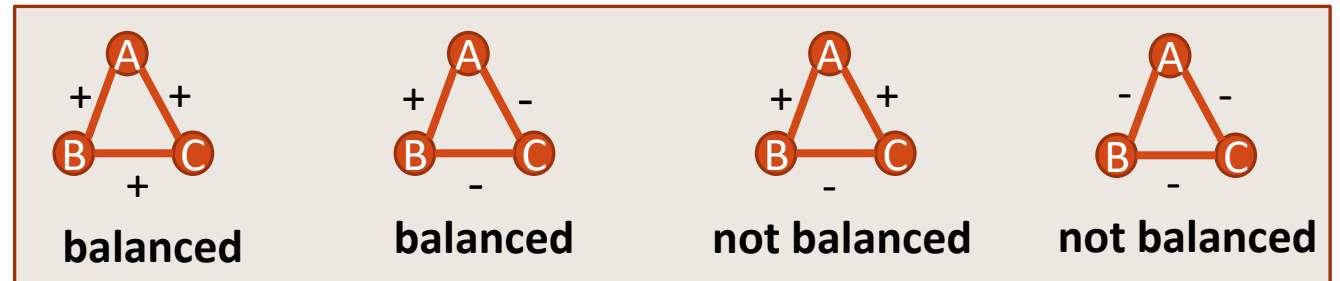


not balanced

Positive and Negative Relationships: Balance Theorem

T: If a labeled complete graph is balanced, then either

- all pairs of nodes are friends, or
- the nodes can be divided into two groups, X and Y , such that every pair of nodes in X like each other, every pair of nodes in Y like each other, and everyone in X is the enemy of everyone in Y .



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Proof:

- Let N be a balanced network
- If all edges are positive – O.K.
- If there is at least one negative edge, pick a node A
 - $X :=$ all friends of A
 - $Y :=$ all enemies of A

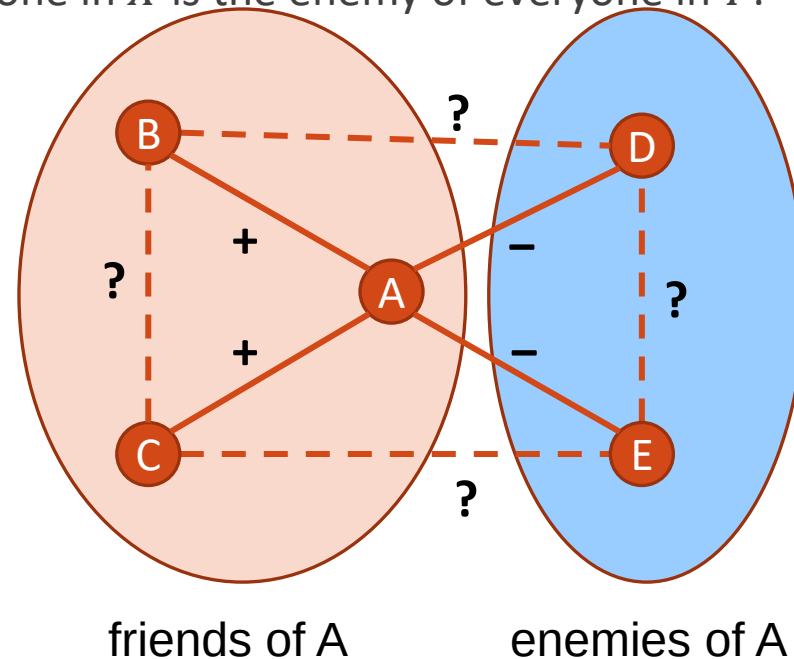
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Proof:

- $X :=$ all friends of A (including A)
 - $Y :=$ all enemies of A
1. Every two nodes in X are friends.
 2. Every two nodes in Y are friends.
 3. Every node in X is an enemy of every node in Y .



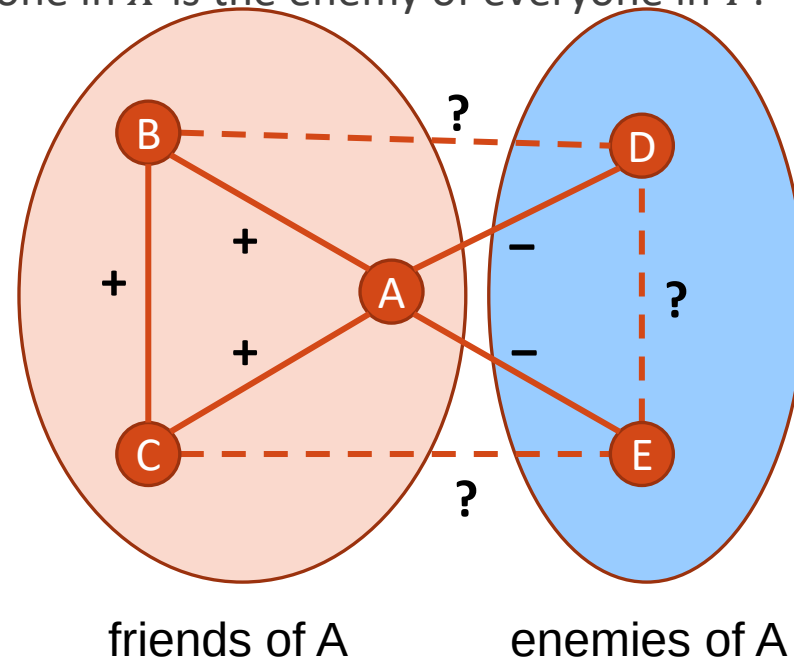
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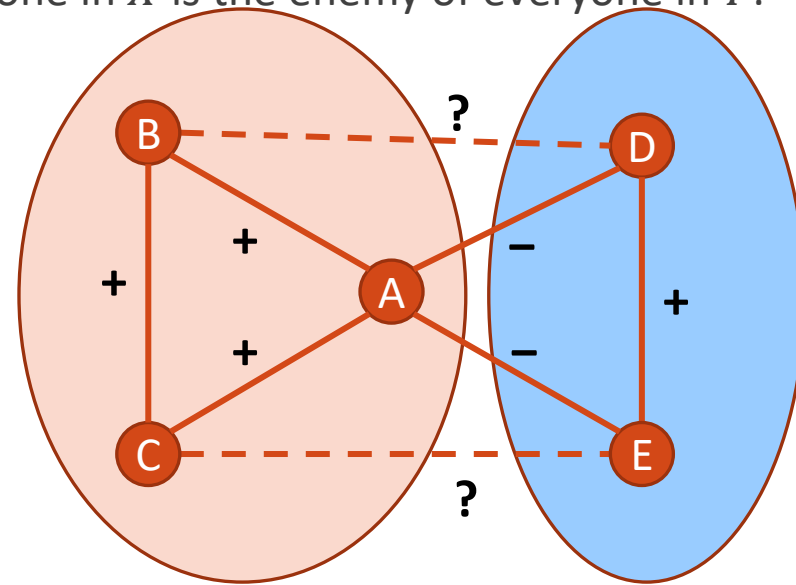
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X – friends of A

Y – enemies of A

Positive and Negative Relationships: Balance Theorem

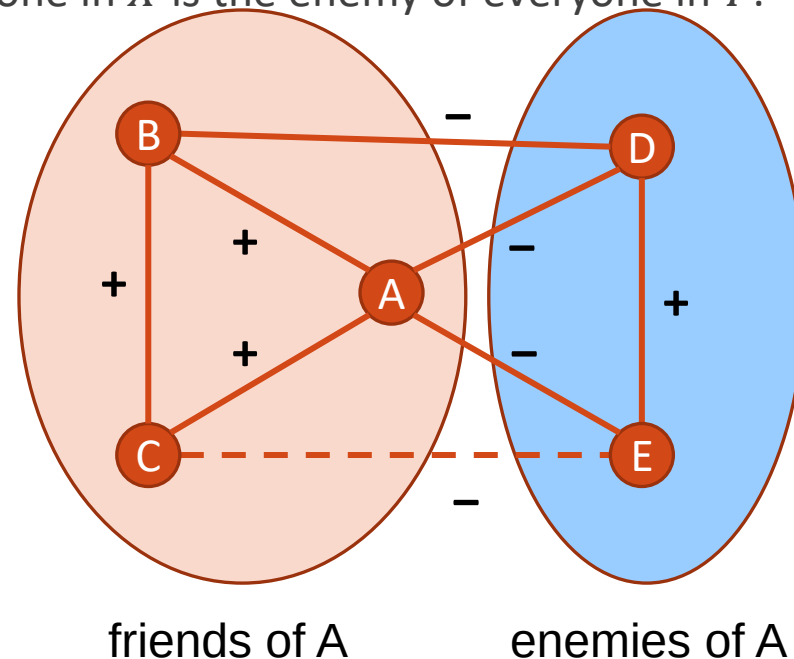
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QED.

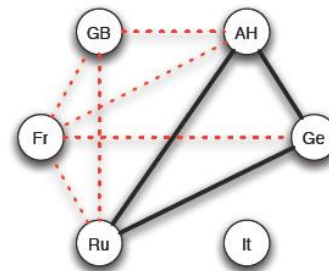


Applications of Structural Balance

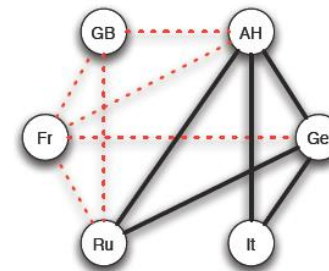
The evolution of alliances in Europe, 1872-1907

- Solid dark edges indicate friendship
- dotted red edges indicate enmity.
- Note how the network slides into a balanced labeling – and into World War I.

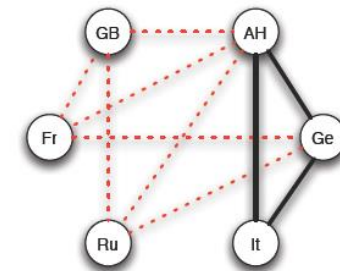
GB = Great Britain
Fr = France
Ru = Russia
It = Italy,
G= Germany
AH = Austria-Hungary



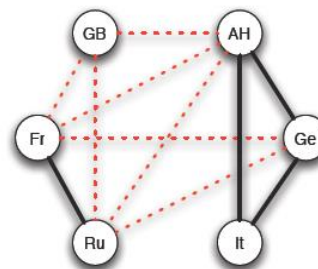
Three Emperors'
League 1872-81



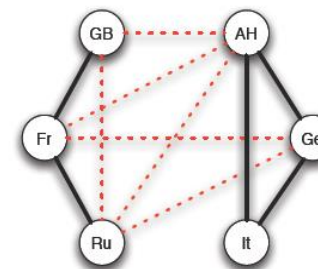
Triple alliance 1882



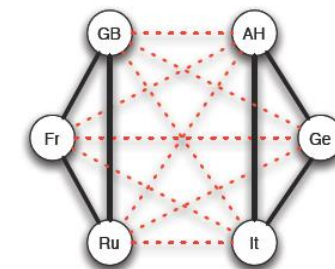
German-Russian
Lapse 1890



French-Russian
Alliance 1891-94



Entente Cordiale
1904



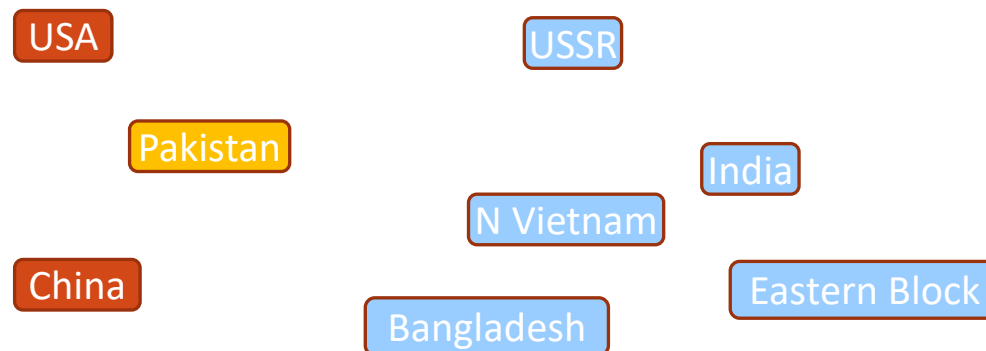
British Russian
Alliance 1907

Figure from <http://www.cs.cornell.edu/home/kleinber/networks-book/networks-book-ch05.pdf>

Applications of Structural Balance

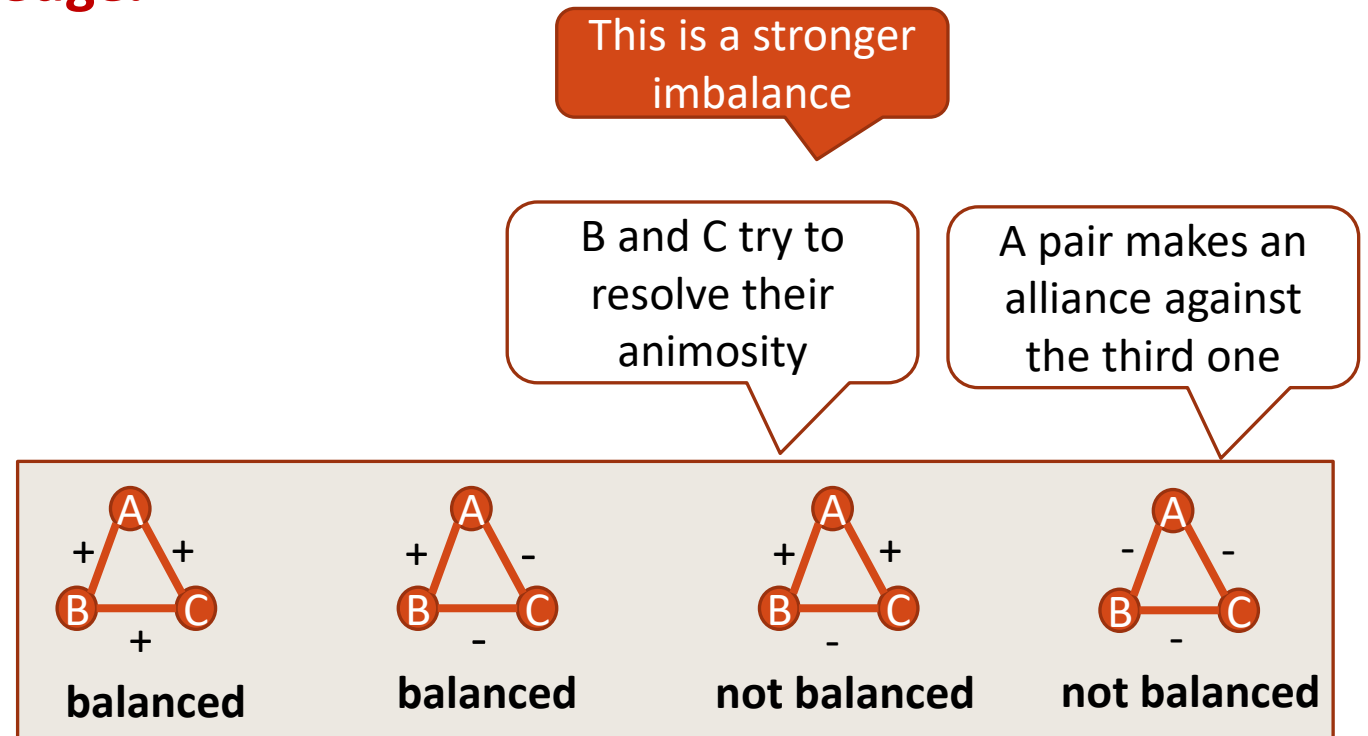
Moore described the conflict over Bangladesh's separation from Pakistan in 1972:

"[T]he United States's somewhat surprising support of Pakistan ... becomes less surprising when one considers that the USSR was China's enemy, China was India's foe, and India had traditionally bad relations with Pakistan. Since the U.S. was at that time improving its relations with China, it supported the enemies of China's enemies. Further reverberations of this strange political constellation became inevitable: North Vietnam made friendly gestures toward India, Pakistan severed diplomatic relations with those countries of the Eastern Block which recognized Bangladesh, and China vetoed the acceptance of Bangladesh into the U.N."



Weak Structural Balance Property

No set of three nodes such that the edges among them consist of exactly two positive edges and one negative edge.



Weak Structural Balance Property

No set of three nodes such that the edges among them consist of exactly two positive edges and one negative edge.

T: If a labeled complete graph is weakly balanced, then its nodes can be divided into groups of mutual friends and every two nodes belonging to different groups are enemies.

Proof:

1. Let A be a node and X is the set of friends of A , including A .
2. All A 's friends are friends with each other.
3. A and all its friends are enemies with everyone else.
4. Remove X from the network and continue with 1.

Generalizing Structural Balance

1. Not all pairs in a network are connected
2. Not all triangles must be balanced, e.g., if **most** triangles are balanced, then the network can be **approximately** divided into two fractions?

Structural Balance in Arbitrary (Non-Complete) Networks

- A. a (non-complete) graph is balanced if it can be “completed” by adding edges to form a signed complete graph that is balanced.
- B. Not all triangles must be balanced, e.g., if **most** triangles are balanced, then the network can be **approximately** divided into two fractions?